NLP 245 Project Report*

Due: March 10th, 2026 11:00AM (before class)

1 Overview

Each team submits one project report worth 20% of your overall grade. The report should be no more than eight pages (excluding references and appendix); otherwise, we will discount the grade. Each group is encouraged to produce a high-quality paper, similar to those submitted at top NLP conferences like ACL or EMNLP workshops. This document outlines the information we anticipate you will include in your report. In addition, we provide a LaTeX template at `https://www.overleaf.com/read/bwqyvwrvddtd#cd4b18`.

2 Required Group Meetings

We are **requiring meetings** with the course staff. These will be held on:

- Tuesday February 24th, 12:35 - 1:15 PM, Room 2235
- Tuesday February 24th, 3:15 - 4:00 PM, Room 2235
- Wednesday March 4th, 3:30 - 5:00 PM, Room 3212
- Thursday March 5th, 3:15 - 4:30 PM, Room 2235

Each group must attend one of these sessions, with all group members present. Please note that the Wednesday section is canceled on February 25th..

3 Report Contents (maximum 8 pages)

Your project report should be written in the style of an NLP research paper. It should be written such that an NLP 245 student can understand the report. Your report should be 8 pages long excluding references. You should make sure to incorporate any feedback you received in your proposal and/or meetings you've had with the course staff. Note that longer is not always better; clear, concise writing will be rewarded. Consider using the section structure below, though you can use other structures.

*Some content is from Stanford CS224n.

3.1 Abstract

An abstract should motivate the problem, describe your goals, describe your contributions, and highlight your main insights/findings in fewer than 300 words.

3.2 Introduction

The introduction describes the problem, why it is difficult, interesting, or important, how and why current methods succeed or fail at solving the problem, and describe the key ideas of your approach and results. Though an introduction covers similar content as an abstract, it offers more space for motivation, detail, references to previous work, to draw the reader's attention.

3.3 Related Work

This section provides an overview of existing work in the field to help the reader understand the research context of your work.

- You could talk about papers that inspired your approach, papers that serve as baselines, papers that propose alternative approaches to the problem, papers that apply your methods to different tasks.
- This section should not go into too many details in any one paper (for example, there should probably be no equations) but rather explain how the papers relate to each other and your project.
- As you review the literature, try to identify limitations or motivations that point to why your work is an exciting next step or promising analysis (or otherwise, if your work does not fall into these categories).

3.4 Approach

This section describes your approach(es) to the problem. For example, this is where you could describe your agent architecture or novel algorithm. Essentially, in this section you are describing what makes your project unique. This section may also go into detail about the technical tools you'll be using in your experiments (e.g., probing methods, proof techniques).

- When describing your main approaches, you should be specific; you should probably include equations and figures.
- You should also describe your baselines. You may do this in detail or refer to another paper for the details, depending on space constraints and how standard your baseline is.
- Make it clear if any part of your approach is original (so we can give you credit). Provide references for models and techniques that are not your own.

- Please provide references or links if you use code from other projects. Also, make it clear when describing something you coded yourself (so we can give you credit).
- Before you begin writing, agree on a fixed technical vocabulary (you defined or is well defined in the literature) and use it consistently throughout the report. This will make it easier for readers to follow and is good practice for general research writing.

3.5 Experiments

This section contains the following.

- **Data:** Describe the dataset(s) you're using (provide references). Make sure the associated task is clearly described. Being precise about the exact form of the input and output can help readers understand your work, especially if you've defined your own task.
- **Evaluation Methods:** Describe the evaluation metric(s) you use, plus any other details necessary to understand your evaluation. Some projects will have clear metrics from prior work on given datasets, but other projects will probably define their metrics. If you're defining your own metrics, be clear about what you're measuring with each evaluation method (whether quantitative or qualitative, automatic or human-defined) and how this is defined.
- **Experimental Details:** Describe how you conducted your experiments (e.g., sampling hyper-parameters, model checkpoints, etc.)
- **Results:** Report the quantitative results you have found. To compare results and compare against baselines, use a table or plot.
  - Please provide a discussion on your quantitative results. Are they expected? Better than your expectation? Worse than your expectation? Why? What does this imply about your approach?

3.6 Analysis

Your report should include a qualitative evaluation. That is to examine key characteristics or outputs to help you understand your system or method (e.g., how it works, when it is effective, and when it fails).

- Types of qualitative evaluation: Commenting on selected examples, error analysis, measuring the performance metric for specific subsets of data, ablation studies, comparing the behaviors of two methods beyond just the metric, and visualizing attention distributions.

3.7 Conclusion

Summarize your project's main findings and lessons learned. Highlight your achievements while noting your work's key limitations. Finally, describe the potential future work.

3.8 References

Your references section should be produced using BibTeX.

3.9 Appendix (optional)

An appendix can also be included, but it must be part of the main PDF and does not count against the 8-page limit. Appendices can help provide additional details, examples, figures, results, visualizations, etc, that may not fit into the main paper. On the other hand, your grader is not required to read the appendix, and you should expect that you will be graded based on the content of the main body of your paper.

4 Improving Your Technical Writing

Here are some resources and tips which may help improve your technical writing.

- Always be precise, and use consistent technical terminology, and define terms that are obvious to you but are unfamiliar to other NLP 245 students. Please also **make sure to cite every source you use and every claim you make that isn't backed up by your own experiments**.
- Tips for Writing Technical Papers, Jennifer Widom.
- Writing in the Sciences, Coursera course.
- Look carefully at multiple NLP papers to understand the typical writing structure, style, and content of different sections. Build your writing model on these examples.
- Revisit the NLP papers you've already read (for example, the paper summary in your proposal). Which parts are easy or difficult to follow and why? Can you figure out any good writing practices that you might use for your writing?
- Look back at your writing from the proposal and milestone. You can likely base a decent amount of your final paper on your initial writing.
- Ask a friend read your writing and tell you whether it is clear. This can be helpful even if your friends do not have the relevant technical backgrounds.
- The course staff is happy to discuss best practices of technical writing. You can bring a draft to office hours to get feedback.

5 Grading and Feedback

- Your project report will be evaluated holistically, taking into consideration several aspects such as novelty, performance of your methods, complexity of the underlying techniques, thoroughness of your evaluation, amount of work put into the project, analysis quality, writing quality, demonstrating insights etc.
- All the course staff will grade your report. The scores will be averaged to determine your final grade. In addition, you will receive feedback on your report.

6 Submission Instructions

Submit your PDF on Canvas as before.